



TAYLOR'S UNIVERSITY
FEBRUARY 2026 SEMESTER

**Machine Learning and Intelligent System
(CSC61304)**

**Assignment 3 – Final Individual Project (40%)
Title: Advanced Intelligent System Design**

DUE DATE: End of Week 7 (Before End of Semester – 20 March)

Student Declaration				
1. I confirm that I am aware of the University's Regulation Governing Cheating in a University Test and Assignment and of the guidance issued by the School of Computing and IT concerning plagiarism and proper academic practice and that the assessed work now submitted is in accordance with this regulation and guidance. 2. I understand that, unless already agreed with the School of Computing and IT, assessed work may not be submitted that has previously been submitted, either in whole or in part, at this or any other institution. 3. I recognize that should evidence emerge that my work fails to comply with either of the above declarations, then I may be liable to proceedings under Regulation.				
Student Name	Student ID	Date	Signature	Score
Angelica Julie Herliman	0376752	14-16 March 2026		

Table of Contents

<i>Problem Definition</i>	4
1.1 Real World Problem.....	4
1.2 Importance Problem	4
1.3 Stakeholder	4
1.4 Input Data and Expected Output.....	5
<i>System Architecture Design</i>	6
2.1 System Architecture	6
2.2 Data Flow	7
2.3 Assumptions	7
2.4 Limitations.....	7
<i>Data & Preprocessing</i>	8
3.1 Data Sources.....	8
3.2 Cleaning Strategy	8
3.3 Feature Selection	11
3.4 Train-Test Split.....	12
<i>Model Development</i>	13
<i>Evaluation & Performance Analysis</i>	15
5.1 Confusion Matrix	15
5.2 Evaluation Metrics	16
5.3 Visualization	17
5.4 Result	19
5.5 Strengthens and Weakness	19
<i>Ethics & Responsible AI</i>	21
6.1 Bias Risk	21
6.2 Misuse Risk.....	21
6.3 Data Fairness	21
6.4 Limitations.....	22
<i>Critical Reflection</i>	23
7.1 What Worked Well	23
7.2 Challenges	23
7.3 Next Improvements	23

7.4 Relatation to Intelligent System.....	24
<i>References</i>	25
<i>Appendix</i>	26

Problem Definition

1.1 Real World Problem

The telecommunication industry is rapidly growing and becoming increasingly competitive. Many companies promote and provide similar products but with their own specifications or unique features. This becomes a threat for each telecommunication company because often they lose customers since they stop using the service and move to competitors.

If this problem is not addressed properly, it can become a serious threat to the company. If the company cannot identify churn early, the company will face higher marketing costs to acquire new customers as well as a decrease in revenue which may negatively affect the company's cash flow.

Therefore, one way to overcome this is that the company can use an intelligent system based on machine learning. This system aims to analyse patterns of customer behaviour based on service usage data and customer information. By using machine learning, the system can predict which customers are likely to stop subscribing. This system also acts as a decision support system that helps the company make data-based decisions related to customer retention strategies.

1.2 Importance Problem

In the current industrial era with tight competition, customer churn becomes an important problem that must be addressed by the companies because it has significant impact on business performance. Losing customers causes company revenue to decrease and finding new customers is usually much more expensive than retaining existing customers. Churn prediction can help companies to carry out customer retention strategies so that the company can take actions such as offering promotions or improving services.

1.3 Stakeholder

In this system, several important stakeholders are involved:

- **Telecommunication company management**
Company management who use the results from the system to understand the level of customer loss and make strategic decisions
- **Marketing department**
The marketing department uses information from the system to design more targeted customer retention strategies such as giving special promotions, discounts, or attractive service package offers.
- **Customer Relationship Management (CRM) Team**
The CRM team plays a direct role in maintaining relationships with customers. They use churn predictions to identify customers who have the potential to stop subscribing earlier and then take proactive actions such as contacting customers, responding to complaints, and offering solutions.
- **Customer**
Customers are also parties affected by this system because the company's decisions taken based on the analysis can influence their service experience.

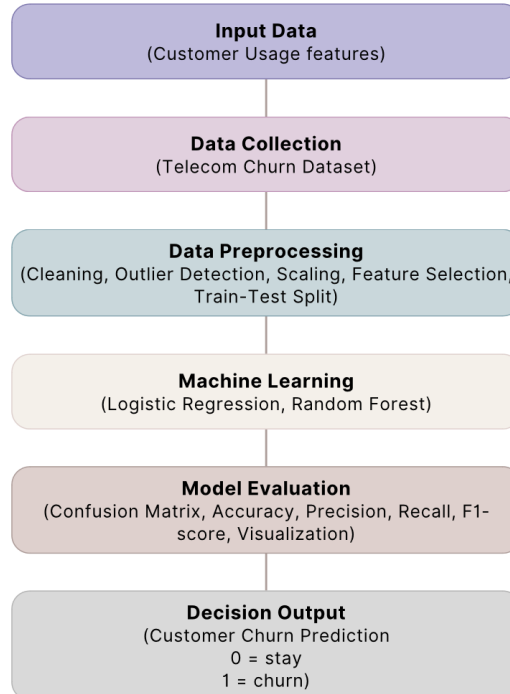
1.4 Input Data and Expected Output

The system inputs that come from telecommunication customer data that will be used are AccountWeeks (new customers are more at risk of churn), ContractRenewal (customers without contracts are easier to churn), CustServCalls (many complaints can increase churn risk), DayMins (high usage often causes churn), MonthlyCharge (high costs cause churn), OverageFee (additional charges make customers dissatisfied which can increase churn risk), and DataUsage (internet usage behaviour that may influence customer satisfaction and churn).

From these inputs, the output is a prediction whether the customer will churn or not. The expected output is binary (1–0) where 0 means customer stays and 1 means customer churns. This prediction can later be used by the company to make business decisions.

System Architecture Design

2.1 System Architecture



The first component in the system is the input data, which is telecommunication customer data used as the basis in the churn prediction process. The data used includes several important attributes such as Account weeks, Contract renewal, Customer service calls, Day mins, Monthly Charge, Overage Fee, and Data Usage. These attributes represent service usage behavior, the costs paid by customers, and customer interactions with customer service.

The second stage is where customer data is collected from the Telecom Customer Churn dataset obtained from Kaggle. This dataset contains information about customer service usage behavior that can be used to analyze the possibility of customers stopping their subscription. The collected data will then be stored in the form of a dataset that is ready to be processed further in the machine learning system.

After the data is collected, the data must first be cleaned through data preprocessing to improve data quality so that the machine learning model can work optimally. Several preprocessing steps carried out include checking missing values, checking duplicate records, outlier detection using the IQR method, feature scaling using StandardScaler and MinMaxScaler, feature selection to choose the most relevant features, and Train-Test Split. This stage is important to ensure that the data used in training the model has good quality and does not contain many errors that can reduce the prediction accuracy level.

The next stage is using a machine learning model to learn patterns from customer data. This study uses two classification algorithms, namely Logistic Regression and Random Forest. The model is

trained using the training dataset so that it can learn the relationship between customer features and the patterns that may lead to churn.

After the model is trained, the model performance is then evaluated to measure how well the model predicts customer churn. The evaluation metrics used are classification metrics such as confusion matrix, accuracy, precision, recall, and F1-score, as well as visualization to make the interpretation and understanding of the model results easier. With this evaluation, it becomes easier to determine which model has the best performance in prediction.

The final stage is the decision output, which is the prediction result from the machine learning model. The system output of the churn prediction is in the form of binary classification, where 0 means the customer continues subscribing and 1 means the customer churns. These prediction results can be used by the company as a data-based decision support system to help determine customer retention strategies such as providing promotions, offering discounted service packages, or improving service quality.

2.2 Data Flow

The workflow in this system begins with collecting customer data which is then cleaned and organized through the preprocessing stage. After the data is clean and ready to be used, the data is used to train the machine learning model. The trained model will then be evaluated using classification metrics to see how accurate the predictions are. The evaluation results will be used to generate churn predictions that will help the company make business decisions.

2.3 Assumptions

Several assumptions used in this design are that the customer data used represents customer behavior accurately, the selected features have an influence on the possibility of customer churn, and the training and testing data have similar distributions so that the model can generalize well.

2.4 Limitations

Several limitations in this system are that the dataset used has a limited number of features so it may not cover all factors that influence customer churn, the machine learning model depends on the quality of the available data, and the predictions produced are probabilistic, so they do not always guarantee correct results in every case.

Data & Preprocessing

3.1 Data Sources

In this study, telecommunication data called the Telecom Customer Churn Dataset obtained from Kaggle is used. This dataset contains information about telecommunication customers that is used to predict whether customers will stop subscribing to the service or continue using the company's services. The dataset consists of 333 customer records with several service usage and billing attributes.

This dataset has several attributes that play a role in decision making for predicting churn. These attributes are related to customer service usage such as churn, account weeks contract renewal, data plan, data usage, customer service calls, average daytime minutes, average daytime calls, monthly bill, and overage fee. The target variable in this dataset is churn which indicates whether the customer stops subscribing or not.

3.2 Cleaning Strategy

Before using the dataset for prediction, pre-processing needs to be carried out to ensure that the quality of the data used is good so that machine learning can produce accurate prediction results.

```
missing_values = df.isnull().sum()

missing_table = pd.DataFrame({
    "Feature": missing_values.index,
    "Missing Values": missing_values.values
})

missing_table = missing_table[missing_table["Missing Values"] > 0]

if missing_table.empty:
    print("No missing values")
else:
    print(missing_table)
```

No missing values

First, checking for missing values is conducted to ensure that there are no missing values in the dataset. If there are missing values, the data will be analyzed to determine whether it needs to be deleted or corrected.

```
duplicate_count = df.duplicated().sum()

if duplicate_count == 0:
    print("No duplicate records")
else:
    print(f"Number of duplicate rows: {duplicate_count}")
```

No duplicate records

Next, duplicate records are checked to ensure that there is no customer data recorded more than once.


```

numeric_cols = df.drop("Churn", axis=1).select_dtypes(include=np.number).columns

outlier_results = []

for col in numeric_cols:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower_bound) | (df[col] > upper_bound)]

    outlier_results.append({
        "Feature": col,
        "Outlier Count": outliers.shape[0]
    })

outlier_table = pd.DataFrame(outlier_results)

if outlier_table["Outlier Count"].sum() == 0:
    print("No outliers detected")
else:
    display(outlier_table)

```

	Feature	Outlier Count
0	AccountWeeks	18
1	ContractRenewal	323
2	DataPlan	0
3	DataUsage	11
4	CustServCalls	267
5	DayMins	25
6	DayCalls	23
7	MonthlyCharge	34
8	OverageFee	24
9	RoamMins	46

Outlier detection is also performed on numerical features using the Interquartile Range or IQR method (a statistical method used to measure data dispersion and detect outliers (extreme values)). This method helps identify values that are too far from the normal distribution so that they can affect the model performance.

These steps are important to ensure that the data used in training the model has good quality and does not contain errors that can affect the prediction results later.

```

scaler = StandardScaler()

scaled_data = scaler.fit_transform(df[numeric_cols])

scaled_df = pd.DataFrame(scaled_data, columns=numeric_cols)

print("Standardized Data:")
display(scaled_df.head())

```

	AccountWeeks	ContractRenewal	DataPlan	DataUsage	CustServCalls	DayMins	DayCalls	MonthlyCharge	OverageFee	RoamMins
0	0.676489	0.327580	1.617086	1.480204	-0.427932	1.566767	0.476643	1.990727	-0.071584	-0.085008
1	0.149065	0.327580	1.617086	2.266072	-0.427932	-0.333738	1.124503	1.564510	-0.107082	1.240482
2	0.902529	0.327580	-0.618396	-0.641642	-1.188218	1.168304	0.675985	-0.262133	-1.574346	0.703121
3	-0.428590	-3.052685	-0.618396	-0.641642	0.332354	2.196596	-1.466936	0.042307	-2.741846	-1.303026
4	-0.654629	-3.052685	-0.618396	-0.641642	1.092641	-0.240090	0.626149	-0.931902	-1.037927	-0.049184

```

normalizer = MinMaxScaler()

normalized_data = normalizer.fit_transform(df[numeric_cols])

normalized_df = pd.DataFrame(normalized_data, columns=numeric_cols)

print("Normalized Data:")
display(normalized_df.head())

```

Normalized Data:

	AccountWeeks	ContractRenewal	DataPlan	DataUsage	CustServCalls	DayMins	DayCalls	MonthlyCharge	OverageFee	RoamMins
0	0.524793	1.0	1.0	0.500000	0.111111	0.755701	0.666667	0.770812	0.542606	0.500
1	0.438017	1.0	1.0	0.685185	0.111111	0.460661	0.745455	0.698869	0.537658	0.685
2	0.561983	1.0	0.0	0.000000	0.000000	0.693843	0.690909	0.390545	0.333150	0.610
3	0.342975	0.0	0.0	0.000000	0.222222	0.853478	0.430303	0.441932	0.170423	0.330
4	0.305785	0.0	0.0	0.000000	0.333333	0.475200	0.684848	0.277492	0.407916	0.505

Then feature scaling is also carried out to adjust the scale of values in numerical features so that all features have a comparable range of values. In this dataset two scaling techniques are used, namely Standardization (StandardScaler) which functions to transform the data so that it has a mean of 0 and a standard deviation of 1, and Min-Max normalization (MinMaxScaler) which transforms feature values into a range between 0 and 1. This scaling helps improve the performance of machine learning models, especially for algorithms that are sensitive to scale such as Logistic Regression.

3.3 Feature Selection

```
selected_features = [  
    'AccountWeeks',  
    'ContractRenewal',  
    'CustServCalls',  
    'DayMins',  
    'MonthlyCharge',  
    'OverageFee',  
    'RoamMins',  
    'DataUsage'  
]  
  
target = 'Churn'  
  
df_selected = df[selected_features + [target]]  
  
print("\nSelected Features =")  
display(pd.DataFrame(selected_features, columns=["Selected Features"]))  
  
print("\nNew Dataset Shape =")  
print(df_selected.shape)  
  
print("\nSelected Dataset Preview:")  
display(df_selected.head())
```

Selected Features =

Selected Features	
0	AccountWeeks
1	ContractRenewal
2	CustServCalls
3	DayMins
4	MonthlyCharge
5	OverageFee
6	RoamMins
7	DataUsage

New Dataset Shape =
(3333, 9)

Selected Dataset Preview:

	AccountWeeks	ContractRenewal	CustServCalls	DayMins	MonthlyCharge	OverageFee	RoamMins	DataUsage	Churn
0	128	1	1	265.1	89.0	9.87	10.0	2.7	0
1	107	1	1	161.6	82.0	9.78	13.7	3.7	0
2	137	1	0	243.4	52.0	6.06	12.2	0.0	0
3	84	0	2	299.4	57.0	3.10	6.6	0.0	0
4	75	0	3	166.7	41.0	7.42	10.1	0.0	0

After preprocessing is carried out, feature selection is performed to choose the most relevant features in predicting customers. In this study, several features are selected as input variables namely AccountWeeks, ContractRenewal, CustServCalls, DayMins, MonthlyCharge, OverageFee, and DataUsage with the target variable used being churn.

The selection of these features assumes that service usage behavior, monthly costs, and interactions with customer service have an influence on the possibility of customers stopping their subscription. By selecting relevant features, the machine learning model can work more effectively and avoid the use of features that do not provide significant contribution to prediction.

3.4 Train-Test Split

```
X = df_selected.drop("Churn", axis=1)
y = df_selected["Churn"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

split_table = pd.DataFrame({
    "Dataset": ["X_train", "X_test", "y_train", "y_test"],
    "Rows": [X_train.shape[0], X_test.shape[0], y_train.shape[0], y_test.shape[0]],
    "Columns": [X_train.shape[1], X_test.shape[1], "-", "-"]
})

display(split_table)
```

	Dataset	Rows	Columns
0	X_train	2666	8
1	X_test	667	8
2	y_train	2666	-
3	y_test	667	-

After cleaning and scaling are performed, the dataset is then divided into a training set and a testing set with a ratio of 8:2. The training data is used to train the machine learning model so that it can learn patterns from customer data. Meanwhile, the testing data is used to evaluate the performance of the model on data that has never been seen before. The stratified sampling method (a sampling technique where the data is divided into several groups and then samples are taken proportionally from each group) is used to ensure that the distribution of churn and non-churn classes remains balanced in the training and testing datasets.

Model Development

Once the data quality is verified with the help of preprocessing, the machine learning model is created, which predicts whether a customer will terminate the telecommunication services subscription. Two algorithms are employed in this research, which are Logistic Regression and Random Forest. The model is trained on training data which has been split with 80% training and 20% testing data.

The process of training a model is done based on the selected features where the target variable used in the training process is the churn variable. The model is trained with the help of the fit() function on the training data in order to make the model learn the relationship between the customer features and the probability of churn.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler, MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, precision_score, recall_score, f1_score
```

```
log_model = LogisticRegression(max_iter=1000)
```

```
log_model.fit(X_train, y_train)
```

```
y_pred_log = log_model.predict(X_test)
```

The reason behind the selection of Logistic Regression is the fact that it is among the classification algorithms that is typically applied to estimate the likelihood of two classes, churn or not churn. This algorithm operates by estimating the association between independent variables and a probability of an event on a logistic basis.

The benefits of the interpretation of Logistic Regression are that the dependence of the features and the prediction outcome can be explained mathematically. Moreover, many machine learning studies tend to implement Logistic Regression as a baseline model (a simple model to act as a baseline in improving the performance of other models) due to its simple and effective form.

The implementation of the Logistic Regression model is performed through Scikit-Learn library (a Python library of machine learning that is used to create, train, and evaluate machine learning models with ease) and with the parameter `max_iter = 1000`. This parameter defines the maximum number of repetitions of the algorithm used in the process of optimizing the model to identify the model with the best parameters. The iteration value is increased to guarantee that the model will be able to converge particularly in cases whereby the data has multiple numerical variables.

```
rf_model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
  
rf_model.fit(X_train, y_train)  
  
y_pred_rf = rf_model.predict(X_test)
```

The second model that is selected is the Random Forest since this is an ensemble learning algorithm (a set of many machine learning models to make better predictions) where many decision trees are joined together to enhance the prediction accuracy and minimize the chance of overfitting.

Every decision tree of the random forest is trained on a different subset of the data, and the outcomes of the forecasts of all trees are combined to generate the final prediction.

Random Forest can also work with large datasets that contain numerous features and can find the most significant features in making a prediction using the feature importance metric (a metric that demonstrates the extent to which any one feature (input variable) contributes to the model making predictions).

Random Forest is known to generate improved performance than simple classification models due to its capability to predict non-linear associations among data.

Random Forest is implemented using the Scikit-Learn library with the parameter `n_estimators = 100`, which determines the number of decision trees used in the model since a larger number of trees generally improves the stability and predictive performance of the model. The `random_state = 42` to make sure that when the same code is run again the same results of the model can be obtained.

To enhance the quality of analysis, two models are implemented in this research to compare the ability of both algorithms in predicting customer churn. This comparison will be used to determine the best performing model in identifying customers who are at risk of terminating their subscription.

Evaluation & Performance Analysis

5.1 Confusion Matrix

To evaluate the performance of the model in predicting customers, several classification metrics are used, namely Confusion Matrix, Accuracy, Precision, Recall, and F1-Score. These metrics provide a comprehensive overview of the model's ability to perform classification.

```
cm_log = confusion_matrix(y_test, y_pred_log)

cm_log_table = pd.DataFrame(cm_log,
                             index=["Actual No Churn", "Actual Churn"],
                             columns=["Predicted No Churn", "Predicted Churn"])

print("Confusion Matrix")
display(cm_log_table)
```

Confusion Matrix

	Predicted No Churn	Predicted Churn
Actual No Churn	553	17
Actual Churn	77	20

According to the results of the confusion matrix of the Logistic model, the total number of True Negatives is 553, which means that the customers who do not churn are correctly predicted by the model, the total number of True Positives is 20, the total number of False Positives is 17, and the total number of False Negatives is 77. the number of false negatives is relatively high, which means the Logistic Regression model is yet to identify the actual churning customers.

```
cm_rf = confusion_matrix(y_test, y_pred_rf)

cm_rf_table = pd.DataFrame(cm_rf,
                             index=["Actual No Churn", "Actual Churn"],
                             columns=["Predicted No Churn", "Predicted Churn"])

print("Confusion Matrix")
display(cm_rf_table)
```

Confusion Matrix

	Predicted No Churn	Predicted Churn
Actual No Churn	557	13
Actual Churn	36	61

Random Forest on the other hand had an output of 557 True Negatives, 61 True Positives, 13 False Positives and 36 False Negatives. Also, this demonstrates that Random Forest is more effective at identifying churn customers than Logistic Regression.

5.2 Evaluation Metrics

Logistic Regression

```
log_metrics = pd.DataFrame({
    "Metric": ["Accuracy", "Precision", "Recall", "F1-score"],
    "Score": [
        accuracy_score(y_test, y_pred_log),
        precision_score(y_test, y_pred_log),
        recall_score(y_test, y_pred_log),
        f1_score(y_test, y_pred_log)
    ]
})

print("Logistic Regression Performance")
display(log_metrics)
```

Logistic Regression Performance

	Metric	Score	
0	Accuracy	0.859070	
1	Precision	0.540541	
2	Recall	0.206186	
3	F1-score	0.298507	

Random Forest

```
rf_metrics = pd.DataFrame({
    "Metric": ["Accuracy", "Precision", "Recall", "F1-score"],
    "Score": [
        accuracy_score(y_test, y_pred_rf),
        precision_score(y_test, y_pred_rf),
        recall_score(y_test, y_pred_rf),
        f1_score(y_test, y_pred_rf)
    ]
})

print("Random Forest Performance")
display(rf_metrics)
```

Random Forest Performance

	Metric	Score	
0	Accuracy	0.926537	
1	Precision	0.824324	
2	Recall	0.628866	
3	F1-score	0.713450	

Model Comparison

	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.859070	0.540541	0.206186	0.298507
1	Random Forest	0.926537	0.824324	0.628866	0.713450

According to the Model Performance Metrics, it could also be observed that the accuracy of the Random Forest is higher with 92.65% as opposed to Logistic Regression which has a lower accuracy of only 85.91%. This indicates that Random Forest provides better overall classification performance.

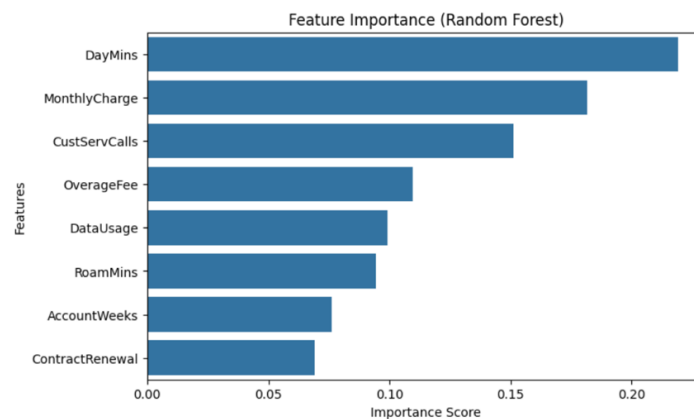
In Precision, Random Forest obtained 0.824 and Logistic Regression obtained 0.541 when the random forest is greater than the logistic regression which implies that the Random Forest produces more accurate churn predictions.

Recall Logistic Regression in Recall is very low 0.206 which shows that this model cannot identify majority of customer churn. Random Forest, conversely, recalls much higher 0.629.

Lastly, the F1-Score of Random Forest is significantly greater by far at 0.713 as opposed to the Logistic Regression at mere 0.299. This F1-Score shows that the balance between precision and recall of the Random Forest is better.

5.3 Visualization

```
importances = rf_model.feature_importances_  
  
feature_names = X.columns  
  
feature_importance_df = pd.DataFrame({  
    "Feature": feature_names,  
    "Importance": importances  
})  
  
feature_importance_df = feature_importance_df.sort_values(by="Importance", ascending=False)  
  
plt.figure(figsize=(8,5))  
sns.barplot(  
    x="Importance",  
    y="Feature",  
    data=feature_importance_df  
)  
  
plt.title("Feature Importance (Random Forest)")  
plt.xlabel("Importance Score")  
plt.ylabel("Features")  
  
plt.show()
```



Visualization is applied to aid in decision making to compare the results of the two models. To begin with, the feature importance derived through the Random Forest model is demonstrated with a bar graph. This graph shows how significant each feature is in deciding by the model in churn detection of customers.

According to the outcomes of the visualization, the traits that impact customer churn the most are DayMins, MonthlyCharge, and CustServCalls. This shows that the number of months using the service, the monthly payments, and the rate at which customers call the customer service are very significant factors that are likely to make a customer discontinue the subscription.

```

cm_log = confusion_matrix(y_test, y_pred_log)
cm_rf = confusion_matrix(y_test, y_pred_rf)

plt.figure(figsize=(5,4))
sns.heatmap(cm_log, annot=True, fmt='d', cmap='Blues')

plt.title("Confusion Matrix - Logistic Regression")
plt.xlabel("Predicted")
plt.ylabel("Actual")

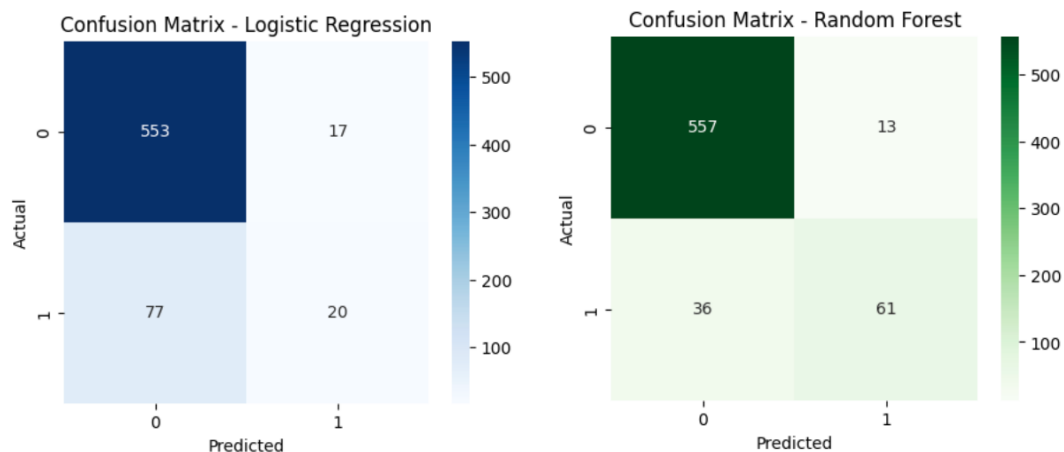
plt.show()

plt.figure(figsize=(5,4))
sns.heatmap(cm_rf, annot=True, fmt='d', cmap='Greens')

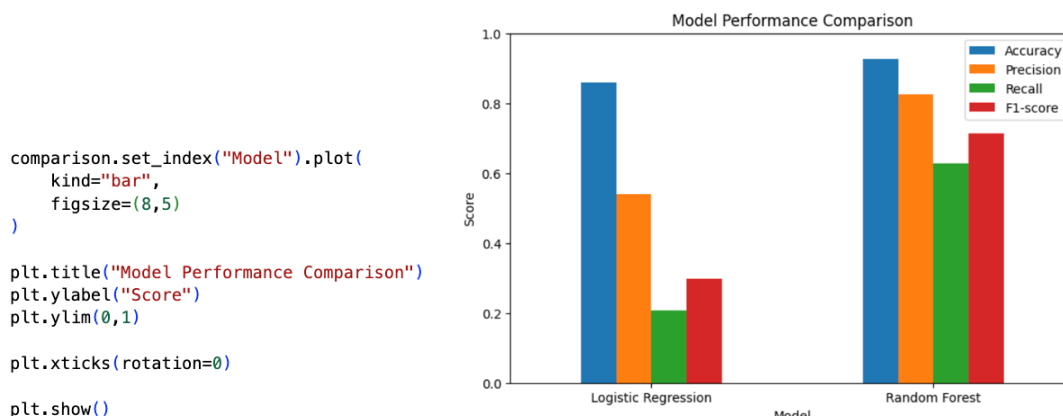
plt.title("Confusion Matrix - Random Forest")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.show()

```



The confusion matrix of both models is also used through visualization in the form of a heatmap. The heatmap gives a visual indication of how many times each model has made the right prediction and how many times the model has made a mistake in classification. Based on the drawing it is possible to note that the number of correct predictions of churn made by Random Forest is higher than the number of correct predictions made by Logistic Regression.



Finally, both models are compared using the metrics of Accuracy, Precision, Recall and F1-Score based on a bar graph. This graph indicates that the values of random forest are always higher on all evaluation metrics than the values of Logistic Regression.

In the comparison of the outcomes of the evaluation, the results of the Random Forest are better than the Logistic Regression when predicting customer churn. This is because the random forest is an ensemble algorithm which uses numerous decision trees to enable it to have more intricate

associations in the data. Conversely, logistic regression has drawbacks of modelling non-linear associations between features.

5.4 Result

In churn prediction, the predictive power of the model to identify the churn potential customers is extremely significant. Hence, a model with a greater recall probability like the Random Forest is more viable to be employed to assist companies to implement customer retention strategies.

Logistic Regression has a few merits in the classification process. The model is not often complicated and can be easily interpreted to explain the connection between the features and the outcomes of the prediction process in a more understandable manner. Moreover, Logistic Regression has low computational complexity thus when training a model, the training process can be done within a short time even when large amount of data is used.

The model may also be applied in cases where the association between the independent variables and the target is relatively linear in nature. Thus, Logistic regression has been frequently adopted as a control model in machine learning studies to give an initial picture of the model performance.

5.5 Strengthens and Weakness

Logistic Regression has several limitations, even though it has benefits. It is a model that has little capacity to capture non-linear relationship among features hence its performance may decline when the data has complex patterns.

According to the results of the model evaluation, the recall value of the Logistic Regression is low, which means that this model is not quite effective at identifying churning customers. This may lead to the failure of the model to identify many churn customers. In churn prediction problems, recall is particularly important because failing to identify customers who are likely to churn may result in missed opportunities for customer retention.

Random Forest is an ensemble learning algorithm that uses many decision trees to enhance the accuracy of prediction. It is a method where the model can model non-linear relationships in the dataset which are complex. The model is capable of good handling of datasets that have high number of features and can minimise the chances of overfitting by randomly sampling the data and features. Also, Random Forest offers data on the significance of the features, which are used to determine which factors affect customer churn the most.

According to the results of the evaluation area, Random Forest reflects a larger number of metrics than Logistic Regression, and this fact means that the given model is more effective in terms of customer churn prediction.

Random Forest is also not without weaknesses even though it has good performance. It is more complex than the previous model thus, the training and prediction tasks of this model take a longer time to compute than that of the Logistic Regression model. Moreover, the fact that it is a combination of numerous decision trees implies that it can be regarded as a black-box model (a model whose inner workings are hard to comprehend or explain) and thus model interpretation is more challenging than with less complex models.

Random Forest is more successful than the Logistic Regression when it comes to predicting customer churn, based on the analysis of the evaluation results and model performance. Random Forest is better to be applied to the datasets with non-linear relationship and complicated interactions between features, customer churn datasets with different factors of service usage and customer expenses.

By considering the performance and characteristics of the model, Random Forest is the more effective model to be used in the customer churn prediction system in this study.

Ethics & Responsible AI

6.1 Bias Risk

Ethical considerations and responsibility in the context of machine learning-based systems are highly critical since the system not only analyses customer data but also influences business decisions that directly affect customers. Hence, some ethical provisions that should be addressed in this research are the threat of bias, data confidentiality, the possibility of abusing the framework, data impartiality, and model constraints.

Bias in the data is one of the primary risks of machine learning that would influence the prediction performance of the model. The bias may take place when the dataset employed does not capture the whole population of customers in an equal manner. Moreover, there can be a class imbalance in the data in the case of churn prediction, where the non-churning customers are significantly more than the churning ones. This situation may cause the model made to predict the majority and neglect those customers who are in fact at risk of churn. This can cause the model to be accurate, but not effective in determining the presence of churn customers and hence making poor business decisions and causing undue bias against some group of people.

To minimize the possibility of bias, one should make sure that the dataset is distributed in a representative way and use procedures like stratified sampling and balancing data methods that will guarantee that the proportions of classes are preserved during the training session. Besides that, feature selection should be well done to eliminate features that thus indirectly bring bias into the model.

The data that will be utilized in a churn prediction system can include sensitive details about the customer, including patterns of using the services, their costs, and customer behavior. Privacy protection is, therefore, very important. The data must also be anonymized and must not include personally identifiable information including name, address, or phone numbers. Moreover, businesses should also make sure that the collection and utilization of data are in line with the relevant data protection policies and are utilized with the purpose that is pertinent to the analysis.

6.2 Misuse Risk

The misuse of the churn prediction system is a potential risk if prediction results are not used appropriately. This may lead to ethical concerns where certain customers are systematically disadvantaged based on model predictions rather than their actual behaviour. For example, companies may disregard customers who are predicted to have a high risk of churning or reduce the quality of services offered to them. Such actions may result in unfair treatment of customers. Therefore, prediction results should be used as a decision support tool rather than the sole basis for decision-making. Human judgment remains essential to ensure that decisions are made within a broader business and ethical context.

6.3 Data Fairness

The element of data equitability is also a feature that should be taken into consideration when developing this system. The model should have the capability to give consistent, and equal predictions to all customers without discriminating against some groups. In case the data is not representative, or imbalanced, then the model can generate biased estimates in the direction of the

majority group. Thus, besides preprocessing, fairness assessment should be conducted to evaluate whether the model performs consistently across different customer groups.

6.4 Limitations

Despite the relatively good performance of the models employed in this research, it also has several limitations that should be addressed. First, the model employs few features only and thus it might not encompass the entire factors which affect customer choices to churn. Second, the behavior of customers is usually complicated and could be affected by external factors that are not present in the data hence the model is not able to capture all the dynamics. Moreover, models such as Random Forest provide better predictive performance but are less interpretable than Logistic Regression, which may reduce transparency and trust in the model's decisions.

Moreover, prediction errors, particularly false negatives, are critical in churn prediction because failing to identify customers who are likely to churn can result in direct financial loss and missed retention opportunities. This may make the company miss out on the chance of executing retention strategies within an appropriate time. Therefore, the prediction results should be used as a supporting tool rather than a standalone decision-making system, and must be combined with business judgment and domain knowledge.

Critical Reflection

7.1 What Worked Well

One of the aspects that worked well in the development of this system is the structured machine learning pipeline, starting from pre-processing to evaluation and model visualization. Stages such as data cleaning, feature selection, and train-test split successfully produced a dataset that is ready to be used for model training.

In addition, the use of two different algorithms (Logistic Regression and Random Forest) allows for an objective comparison of model performance. After conducting the comparison, the evaluation results show that Random Forest is able to provide better performance, especially in recall and F1-score, which are very important aspects in the case of churn prediction.

To help interpret the results of analysis and machine learning predictions, visualizations such as confusion matrix, feature importance, and model comparison are used to make it easier to understand model performance more deeply and to provide insights into the factors that influence customer churn.

7.2 Challenges

One of the main challenges in this study is understanding how to select features that are truly relevant to churn prediction. Although several features have been selected based on domain assumptions, not all features have a significant contribution to the model.

There is also a challenge in handling the trade-off between precision and recall. In churn prediction, recall is very important because failure to detect customers who churn can lead to losses and missed opportunities for the company. However, increasing recall often causes a decrease in precision, so it is important to balance both recall and precision.

Another challenge is the limitation in understanding complex relationships between features. Simple models such as Logistic Regression are not able to capture non-linear patterns in the data, while more complex models such as Random Forest are difficult to interpret.

7.3 Next Improvements

Therefore, if given more time, several improvements can be made to enhance the quality of the system in predicting churn. First, the use of more features from the dataset can be considered to improve model performance, including performing more in-depth feature engineering.

Second, more experimentation with algorithms such as Gradient Boosting or XGBoost can be conducted to compare the performance of more advanced models.

In addition, hyperparameter tuning on the Random Forest model can be performed to optimize model performance, since in this study the parameters used are still default.

Finally, the use of cross-validation techniques can also be applied to obtain more robust model evaluation and reduce the possibility of overfitting.

7.4 Relation to Intelligent System

The system developed in this study directly reflects the basic concepts of intelligent systems, especially in the use of machine learning to perform classification and automated decision making. The models use historical data to recognize patterns of customer behaviour and generate predictions, which is an implementation of the supervised learning concept in intelligent systems.

However, this system also shows that intelligent systems do not always produce definite decisions, but rather operate under conditions of uncertainty. The predictions generated are probabilistic in nature, so they are not always accurate in representing actual customer behaviour. This shows that the results of the system still need to be interpreted carefully and wisely before being used in decision making.

This system also emphasizes that the performance of an intelligent system is highly dependent on data quality, feature representation, as well as pre-processing and evaluation processes. Limitations in the data, such as lack of features or imbalanced distribution, can directly affect the model's ability to produce accurate and fair predictions.

Furthermore, this system highlights the trade-off between performance and interpretability. Models such as Random Forest are able to provide higher performance but have a lower level of interpretability compared to Logistic Regression. This becomes an important consideration in the implementation of intelligent systems, especially when transparency and trust in the system are crucial factors.

This system can be categorized as a decision support system, where prediction results are used to assist business decision making rather than replace human roles in decision making. This reflects an important principle in intelligent systems that technology should be used to support human decision making, not replace it. Therefore, the integration between model predictions and human judgment becomes key in producing more accurate and responsible decisions.

References

Dataset:

<https://www.kaggle.com/datasets/barun2104/telecom-churn>

- Rigatti, S. J. (2017). Random forest. *Journal of insurance medicine*, 47(1), 31-39.
<https://meridian.allenpress.com/jim/article/47/1/31/131479>
- Hosmer Jr, D. W., Lemeshow, S., & Sturdivant, R. X. (2013). *Applied logistic regression*. John Wiley & Sons.
https://books.google.com/books?hl=en&lr=&id=bRoxQBIZRd4C&oi=fnd&pg=PR13&dq=logistic+regression&ots=kM7NyndUh7&sig=DI0fXgkUzpiYDXAV_uXqt_YU8mc
- García, S., Ramírez-Gallego, S., Luengo, J., Benítez, J. M., & Herrera, F. (2016). Big data preprocessing: methods and prospects. *Big data analytics*, 1(1), 9.
<https://link.springer.com/article/10.1186/s41044-016-0014-0>
- García, S., Luengo, J., & Herrera, F. (2015). Data preprocessing in data mining (Vol. 72, pp. 59-139). Cham, Switzerland: Springer International Publishing.
<https://link.springer.com/content/pdf/10.1007/978-3-319-10247-4.pdf>
- Huang, B., Kechadi, M. T., & Buckley, B. (2012). Customer churn prediction in telecommunications. *Expert Systems with Applications*, 39(1), 1414-1425.
https://www.sciencedirect.com/science/article/pii/S0957417411011353?casa_token=ainybowMozkAAAAA:kGSS3e7PnMzy6y_r0LSl9ncCX2X2boyxy26vRIIdDGeCPUWg-M3wxh6Zqtk4QdYEUaSBviTkozGrqkA
- Lalwani, P., Mishra, M. K., Chadha, J. S., & Sethi, P. (2022). Customer churn prediction system: a machine learning approach. *Computing*, 104(2), 271-294.
<https://link.springer.com/article/10.1007/S00607-021-00908-Y>
- Wilamowski, B. M., & Irwin, J. D. (Eds.). (2018). *Intelligent systems*. CRC press.
<https://books.google.com/books?hl=en&lr=&id=NGfMBQAAQBAJ&oi=fnd&pg=PP1&dq=intelligent+system&ots=d-avnbFKiy&sig=9ugi9xew1EEyRYLe5a6gC4epquM>